

Paragraphs that argue

Turning a Results draft from description into evidence

YOU LEAVE WITH

A Results section where every paragraph makes a claim, backs it with numbers, and links to the next.



Tutorial 6 of 8 • Semester 1

Today's plan

Goal: Turn your Results draft into paragraphs that argue, not describe.

- 1 Methods: reporting a correlation in full; asymmetry
- 2 One paragraph, one job; the order of an argument
- 3 Flow: old information → new information
- 4 Numbers a sceptic would accept

BRING / HAVE READY

- Task 5B Results draft (printed is best: you'll mark it up)
- Highlighters or coloured pens
- Your figures and captions



METHODS

Reporting a correlation in full

- A correlation in a report needs: **r**, **n** (and **n_eff**), **p** or a 95% range, the **period**, and **what** was correlated with what.
- Add the **slope** for the size of the effect, with units (pp per °C).
- Two decimal places for **r**; '≈' for **n_eff**.
- 'Significant' only if you tested it, and say against what.
- Notebook \$6 prints every number you need: **r**, **n**, **n_eff**, **r_lo95**, **r_hi95**, **slope_per_ONI**, **p_adj**.

TEMPLATE (ILLUSTRATIVE NUMBERS)

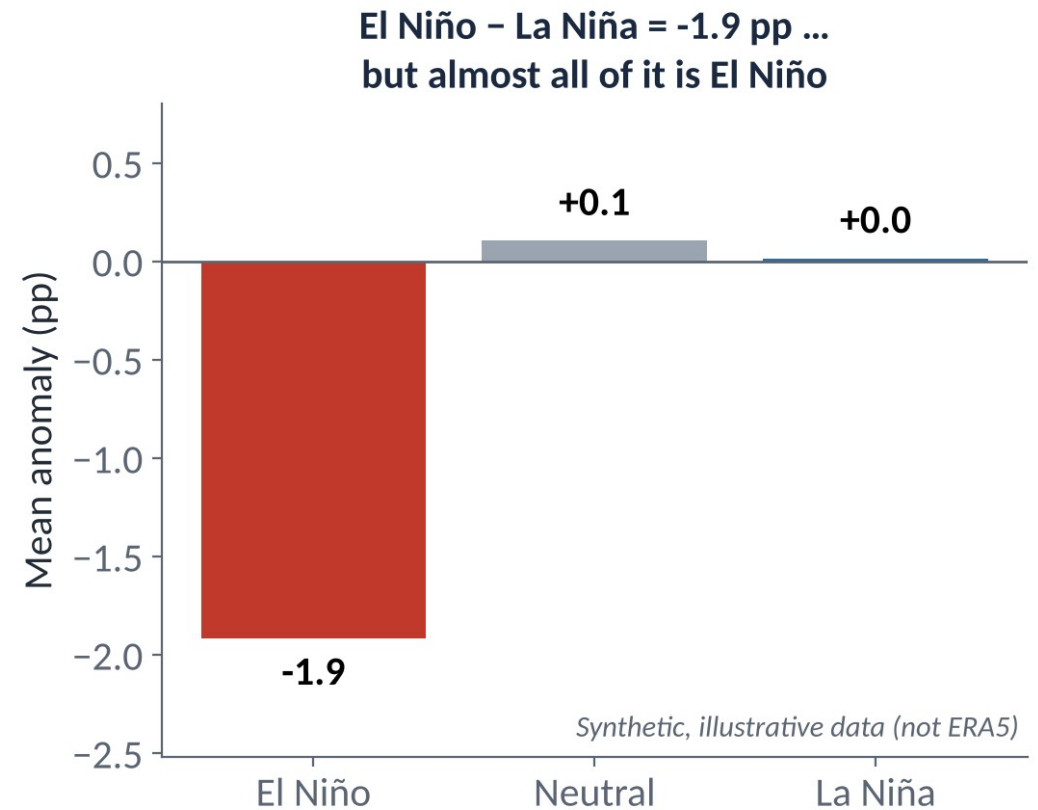
“Monthly high-cloud anomalies over [region] are negatively correlated with the ONI ($r = -0.45$, 95% range -0.63 to -0.22 ; $n = 247$ months, $n_{\text{eff}} \approx 60$; $p < 0.001$; January 2005 – July 2025), a decrease of about 2.1 pp per °C of ONI.”

Every number earns its place: strength, uncertainty, sample, significance, period, size.

METHODS

El Niño and La Niña are not mirror images

- The atmosphere often responds more strongly, or in a different place, to one phase than the other.
- Both the El Niño – La Niña difference and r **hide** this.
- Compare **each phase against neutral**. The notebook prints mean anomalies for El Niño, Neutral and La Niña months (sections 3 and 4).
- A one-sided response is a strong Results sentence **and** a great Discussion point.



IN YOUR RESULTS, THIS BECOMES

“The composite difference (–1.9 pp) comes almost entirely from El Niño months; La Niña months are close to neutral.”

One paragraph, one job

- Schultz: paragraphs are the ‘fundamental organisational unit’ of a paper. Each has **unity** (one theme) and **coherence** (sentences in a logical order).
- The **topic sentence** announces the job, usually first.
- Two themes in one paragraph? **Split** it, **widen** the topic sentence, or **delete** one theme.
- Aim for **4–8 sentences**; avoid one-sentence paragraphs.
- Speed-readers read first sentences (proposal tutorials). If your point is in sentence 5, many readers miss it.

ANATOMY OF A RESULTS PARAGRAPH

Topic sentence: the claim

Evidence: numbers, figure cited

Evidence / comparison

Qualification: where it doesn't hold

Link onward (optional)

~110–140 words

The order of an argument



SCHULTZ'S WARNING

- He gives a paragraph that opens with 'We speculate that...' and only then shows the data.
- Readers become **sceptical**: they've been asked to believe before seeing the evidence.
- Rewritten in the order data → results → interpretation → inference → speculation, the **same sentences** persuade.
- Flag speculation honestly: "We speculate that...", "One possible explanation is..."

SO WHICH COMES FIRST?

- **Assert the finding first** (topic sentence), then show the evidence: that's assert-justify, and it fits this chain.
- **Never lead with speculation.**
- It maps onto the handbook's pattern: **Claim → Evidence → Interpretation → Qualification.**
- In a short report, a light interpretation can follow a result; deeper mechanisms go in the Discussion.

From a list of numbers to an argument

A LIST

“The correlation between high-cloud anomaly and ONI over CP is 0.8. The El Niño composite is +14. The La Niña composite is −12. Over MC the correlation is −0.6. High cloud decreases in El Niño over MC. Some years do not follow this.”

AN ARGUMENT

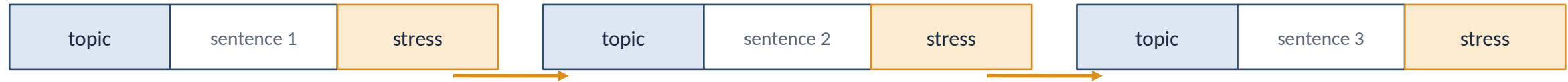
“High cloud responds to ENSO in opposite senses on either side of the Pacific warm pool (Figs 2–3). Over CP, high-cloud anomalies are positively correlated with the ONI ($r = 0.8$, $n = 247$ months); consistently, El Niño months average 14 percentage points above normal, whereas La Niña months average 12 points below. In contrast, high cloud over MC is negatively correlated with the ONI ($r = -0.6$). The MC relationship is weaker, however: [n] of [N] El Niño events show little response there.” (illustrative)

WHAT CHANGED

- A **topic sentence** states the claim; transitions (**consistently, whereas, in contrast, however**) show how each fact relates to the last.
- Numbers gain **units and sample size**: ‘percentage points’ and ‘ $n = 247$ months’, not a bare ‘+14’.
- The exception becomes a **qualification** with a count, not a vague ‘some years’.

Flow: old information first, new information last

Gopen & Swan (via Schultz): the start of a sentence (topic position) holds information the reader already has; the end (stress position) holds what's new and important. The stress of one sentence becomes the topic of the next.



CHOPPY

“ONI is a three-month running mean of SST anomalies in the Niño 3.4 region. Composites were constructed using a ± 0.5 °C threshold. Cloud anomalies were averaged for these months.”

LINKED

*“ENSO is represented by the Oceanic Niño Index (ONI), a three-month running mean of SST anomaly in the **Niño 3.4 region**. **Months** with ONI above $+0.5$ °C are classed as El Niño, and months below -0.5 °C as **La Niña**. **Cloud anomalies** are then averaged over each set of months to form the El Niño and La Niña composites.”*

Numbers a sceptic would accept: spot the problem

SENTENCE A

“Cloud cover increased by 10% during El Niño.”

Percent or **percentage points**? Relative to what? Where, and in which months?

SENTENCE B

“The correlation was $r = 0.4837261$.”

False precision. Give 2 d.p. Add the **period**, **n** and what was correlated with what.

SENTENCE C

“El Niño significantly reduces cloud over the Maritime Continent.”

Was a test done? If not, drop ‘significantly’. ‘Reduces’ is causal: say ‘is associated with lower...’ unless you argue the mechanism.

SENTENCE D

“Figure 4 clearly shows a strong relationship.”

Describe the science, not the figure. ‘Clearly’ and ‘strong’ are unquantified: give r , the slope, or the size of the difference.

IN-CLASS ACTIVITY • 16 MIN

Paragraph surgery on your own Results

- 1 **Highlight the first sentence** of each Results paragraph. Read only those, in order. Do they answer your question?
- 2 Label every sentence **C / E / I / Q** (claim, evidence, interpretation, qualification). Move deep interpretation to a 'Discussion parking lot'.
- 3 **Circle every number.** Does each have units, a baseline, a period or n? Fix percent vs percentage points.
- 4 Find the **weakest link** between two sentences and repair it (repetition, transition or reordering).
- 5 Swap **one paragraph** with your partner, who marks where they got lost.

YOU LEAVE WITH

- A marked-up Results draft and a list of fixes (Task 6A)
- A 'Discussion parking lot': the start of Tutorial 7

PAIR CHECK

- Can you say the paragraph's claim in **one breath** after reading it?
- Is there any sentence that just **describes the plot**?
- Is the exception **counted**?

Before Tutorial 7 • Thu 26 Nov

One week. About 2½ hours. You'll then have Introduction, Methods and Results in draft.

6A • 45 min

Revise Results

- Apply today's surgery to **every** paragraph
- Topic sentences = findings, in order
- Within ~650 words

6B • 60 min

Introduction, full draft ~350 words

- Rewrite 3A **now that you know your results**
- Context → however → purpose (+ hypothesis)
- Fold in your 3B synthesis paragraph

6C • 30 min

Discussion brainstorm

- List **3 possible physical explanations** for your main result
- List **2 limitations that genuinely matter**
- Play devil's advocate: what would a critic say?

NOTEBOOK STRAND

- Only what your Results need: fill any [] numbers; optional sensitivity check.
- No new directions now. Your question is fixed.

READ • SCHULTZ (2009) ELOQUENT SCIENCE

- **Ch. 8** Constructing effective paragraphs
- **§7.3** Structuring logical arguments
- **§9.1** Active vs passive voice